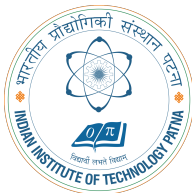


Chapter 08: 802.11ac (Wi-Fi 5) - Gigabit Wi-Fi

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CS6206 Selected Topics in Wireless Networks



Lecture 8 Learning Objectives

By the end of this lecture, you will be able to:

- Explain why 802.11ac is exclusive to 5 GHz band
- Calculate throughput gains from wider channels (80, 160 MHz)
- Describe MU-MIMO (Multi-User MIMO) and its benefits
- Understand 256-QAM modulation and its SNR requirements
- Analyze beamforming improvements in 802.11ac
- Calculate maximum theoretical throughput for various 802.11ac configurations
- Compare Wave 1 vs Wave 2 802.11ac features

802.11ac: The Gigabit Wireless Standard

- **Ratification:** 2013 (Wave 1), 2016 (Wave 2 enhancements)
- **Wi-Fi Generation:** Wi-Fi 5
- **Frequency Band:** 5 GHz only (no 2.4 GHz support)
- **Maximum PHY Rate:** 6.93 Gbps (theoretical, 8×8 MU-MIMO, 160 MHz)
- **Practical Throughput:** 500-1300 Mbps
- **Key Innovations:**
 - ① Wider channels (80, 160 MHz)
 - ② Multi-User MIMO (downlink)
 - ③ 256-QAM modulation
 - ④ Improved beamforming
 - ⑤ More spatial streams (up to 8)

Philosophy: Go big on 5 GHz, leave 2.4 GHz for legacy/coverage

Why 5 GHz Only? The Spectrum Argument

- **2.4 GHz Limitations:**

- Only 83.5 MHz total spectrum
- 3 non-overlapping 20 MHz channels
- Crowded with interference
- Can't support wide channels

- **5 GHz Advantages:**

- ~500 MHz available (more with DFS)
- Up to 25 non-overlapping 20 MHz channels
- Less crowded (initially)
- Can support 80/160 MHz channels

- **Strategic Decision:** Make 5 GHz the performance band, use 2.4 GHz only for compatibility

Consequence: Dual-band APs still needed for legacy 2.4 GHz devices

Wider Channels: 80 and 160 MHz

- **802.11n:** 20/40 MHz channels
- **802.11ac:** 20/40/80/160 MHz channels
- **Throughput Scaling:** Approximately linear with bandwidth

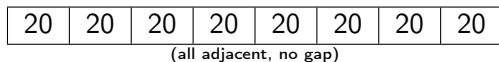
Channel Width	Total Bandwidth	Subcarriers (Data)	Typical Non-overlapping Channels (5 GHz US)	Bandwidth Multiplier
20 MHz	20 MHz	52	~25	1×
40 MHz	40 MHz	108	~12	~2×
80 MHz	80 MHz	234	~6	~4×
160 MHz	160 MHz	468	~3	~8×

Note: 160 MHz can be either contiguous (one 160 MHz block) or 80+80 MHz (two non-adjacent 80 MHz blocks)

Contiguous vs. 80+80 MHz – Visualized

Contiguous 160 MHz

One solid block of 8×20 MHz channels:



80+80 MHz (non-contiguous)

Two separate 80 MHz blocks with a gap:



Why use it?

Allows 160 MHz speed when a single clean 160 MHz block is unavailable (fragmented spectrum).

Problems Caused by the Gap (Not Interference Itself)

The gap **does not create direct interference**, but it introduces **other issues**:

- ➊ **Hardware complexity** – Two independent radio chains are needed (one for each 80 MHz block). Data must be split and recombined.
- ➋ **Different propagation** – The two blocks experience different fading, reflection, and noise levels. Receiver must handle imbalance.
- ➌ **Phase / timing alignment** – Maintaining phase alignment across a gap is harder than in a contiguous block.
- ➍ **Two sources of interference** – Each 80 MHz block can still be jammed by neighbours, radars, or Bluetooth. You are exposed to *twice as many* potential interferers.
- ➎ **Receiver desensitisation** – A strong signal inside the gap can overload the receiver, making it deaf to the wanted 80 MHz blocks.

The gap is not a magic shield

- The gap **avoids** a busy middle part of the spectrum.
- But it **does not eliminate** interference – each block still fights for clean air.
- The extra hardware complexity can cause small delays and higher power consumption.

Simple summary:

“80+80 is like using two separate parking lots far apart. You get the same total space, but you need two drivers, two trucks, and you still have to watch out for other cars in both lots.”

In practice, modern Wi-Fi chips handle this well

What are Subcarriers?

- **Definition:** In OFDM (Orthogonal Frequency Division Multiplexing), a wide frequency channel is divided into multiple smaller, closely spaced **orthogonal** frequencies.
- **The Parallelism Analogy:** If the channel (e.g., 80 MHz) is a multi-lane highway, each subcarrier is a **single lane** carrying its own portion of the data payload.
- **Orthogonality:** Subcarriers are mathematically spaced so they do not interfere with each other, even though they overlap in the frequency domain.

Types of Subcarriers

Not all subcarriers carry user data. They are categorized as:

- **Data Subcarriers:** The workhorses that carry the actual bits of your files/streams.
- **Pilot Subcarriers:** Used for channel estimation and synchronization (keeping the receiver in sync with the transmitter).
- **Guard Subcarriers / Nulls:** Left empty at the edges of the channel to prevent "bleeding" into adjacent channels (interference).
- **DC Subcarrier:** The center frequency, usually left empty to avoid interference from the radio's own electronics.

Why Scale Subcarriers?

- **Throughput:** More subcarriers = more data sent in parallel.
- **Efficiency:** In the table provided in previous slides that 20 MHz has 52 subcarriers, but 40 MHz has 108 (more than double). This is because wider channels can reduce the proportional "waste" of guard bands at the edges.
- **Robustness:** By spreading data across many subcarriers, if one frequency suffers from interference (fading), the rest of the subcarriers can still deliver the message.

Quadrature Amplitude Modulation (QAM)

QAM = Quadrature Amplitude Modulation

Concept:

Converts **digital bits** into **analog waveforms** for wireless transmission.

- **Dual Carriers:** Uses **I** (In-phase, cosine) and **Q** (Quadrature, sine) signals at the same frequency, separated by 90° .
- **Amplitude Variation:** The amplitude of each carrier is adjusted independently.
- **Symbol Mapping:** A single symbol (represented as a point on a 2D constellation grid) conveys multiple bits.

The QAM Mathematical Representation:

$$\text{bits} \rightarrow (I, Q) \rightarrow s(t) = I \cos(2\pi ft) + Q \sin(2\pi ft)$$

Why Not Always Use 256-QAM?

Higher QAM = Higher throughput, but...

- Constellation points get **denser** → smaller distance between points.
- Noise or interference can easily push a point into a neighbour's decision region → **bit errors**.
- Requires **high SNR** (clean signal, close to AP).

Real-world adaptive modulation:

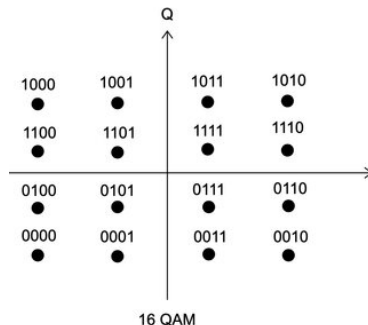
802.11ac/ax automatically switches between BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM based on channel quality.

Constellation Diagram – The Core Idea

Each point = one **symbol** = unique (I,Q) pair.

256-QAM:

- 256 points arranged in a 16×16 grid.
- **8 bits per symbol** (since $2^8 = 256$).
- Example: bits '10110010' map to one specific point.



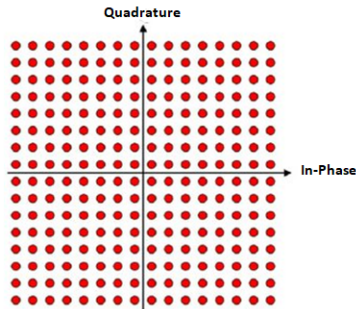
(Conceptual 16-QAM grid; 256 is denser)

Receiver's job: Measure (I,Q), find closest constellation point, map back to bits.

Constellation Diagram – The Core Idea

Each point = one **symbol** = unique (I, Q) pair.

256-QAM:



(Conceptual 16-QAM grid; 256 is denser)

Receiver's job: Measure (I, Q) , find closest constellation point, map back to bits.

OFDM Concept: Divide and Conquer

- **The Core Idea:** Instead of sending a high-speed data stream over one wide channel, OFDM breaks it into many slow-speed streams over multiple **subcarriers**.
- **Parallel Transmission:** It is like moving a large group of people using 50 small cars (subcarriers) simultaneously rather than one giant, slow bus (single carrier).
- **Benefit:** If one "car" hits a traffic jam (interference on one frequency), the other 49 cars still reach the destination safely.

The "Big Hose" vs. "Many Small Pipes"

- **Traditional Way (Single Carrier):** You have one giant pipe (the 20 MHz channel). You pump data through it as fast as possible.
- **The Risk:** If a single "speck of dust" (interference) hits that pipe, the entire stream is corrupted.
- **The OFDM Way (Subcarriers):** You divide that 20 MHz channel into 52 tiny parallel pipes.
- **The Benefit:** You send the data in parallel. If "Pipe 12" gets blocked by noise, Pipes 1–11 and 13–52 keep working perfectly.

The CSE View: Spectrum as an Array

Think of the 20 MHz frequency block as an Array of 52 slots.

- **Mapping:** Your data bits are mapped to these slots:
ChannelArray[0...51].
- **The Process:**
 - 1 You load 52 pieces of data into the array.
 - 2 You transmit the entire array at once (this is one "OFDM Symbol").
 - 3 The receiver reads the array.
- **Subcarriers** are simply the indices of this array in the frequency domain.

What is "Orthogonal"?

In standard physics, if you overlap two radio waves, they create a mess (interference).

- **The "Zip File" Analogy:** Imagine two files overlapping in memory. Usually, that's a crash.
- **The OFDM Trick:** We shape the waves so that when Subcarrier A is at its maximum signal, Subcarrier B is at exactly zero.
- **Result:** They can physically overlap in the air, but the receiver can still pull them apart perfectly using math (the FFT algorithm).

The Magic Number 64 (The Power of 2)

- **FFT Size:** Digital Signal Processors (DSPs) love the "Power of 2." In a 20 MHz channel, we divide the space into 64 potential slots (bins).
- **The Math:** $20 \text{ MHz} \div 64 = 312.5 \text{ kHz}$ per subcarrier.
- **Why not use all 64?** If we used all 64 for data, the signal would "bleed" into the next Wi-Fi channel, causing a massive "memory leak" of interference.

The "Wasted" Slots (Guard Bands)

To keep the signal clean, we intentionally leave some slots empty (Nulls):

- **Guard Subcarriers (11 slots):** We leave the edges of the 20 MHz block empty. This acts like padding in a UI layout, ensuring our data doesn't overlap with the neighbor's channel.
- **The DC Subcarrier (1 slot):** The very center slot (0 Hz offset) is left empty because the radio hardware itself creates a bit of noise there.
- **Remaining:** $64 - 11 - 1 = 52$ active subcarriers.

The Final Breakdown: Data vs. Management

Of those 52 "active" subcarriers, we still can't use all of them for your YouTube stream. We split them into:

- **48 Data Subcarriers:** These carry the actual bits (the payload).
- **4 Pilot Subcarriers:** These are like "heartbeat" signals. They carry known values so the receiver can constantly recalibrate and stay in sync.
- **Analogy:** Think of it as a data frame where you have 52 bytes total, but 4 bytes are reserved for checksums and metadata, leaving 48 bytes for actual data.

Numerical Example 1: Channel Bandwidth Impact I

Problem: A client uses 256-QAM, coding rate $5/6$, and 2 spatial streams. Calculate the PHY rate for:

- ❶ 20 MHz channel
- ❷ 40 MHz channel
- ❸ 80 MHz channel
- ❹ 160 MHz channel

Given:

- 256-QAM: 8 bits per symbol per subcarrier
- Coding rate: $5/6$
- Symbol time: $3.6 \mu s$ (short GI)
- Data subcarriers:
 - 20 MHz: 52
 - 40 MHz: 108

Numerical Example 1: Channel Bandwidth Impact II

- 80 MHz: 234
- 160 MHz: 468

Step 1: Bits per OFDM symbol per spatial stream

$$N_{\text{DBPS}} = N_{\text{SC}} \times N_{\text{BPSC}} \times R$$

- N_{DBPS} = number of **data bits per OFDM symbol**
- N_{SC} = number of **data subcarriers**
- N_{BPSC} = number of **bits per subcarrier**
- R = coding rate
- 20 MHz: $52 \times 8 \times \frac{5}{6} = 346.7$ bits
- 40 MHz: $108 \times 8 \times \frac{5}{6} = 720.0$ bits

Numerical Example 1: Channel Bandwidth Impact III

- 80 MHz: $234 \times 8 \times \frac{5}{6} = 1560.0$ bits
- 160 MHz: $468 \times 8 \times \frac{5}{6} = 3120.0$ bits

Step 2: Symbol rate

$$\text{Symbol rate} = \frac{1}{3.6 \mu s} = 277.78 \times 10^3 \text{ symbols/s}$$

Step 3: PHY rate per spatial stream

$$R = N_{\text{DBPS}} \times \text{Symbol rate}$$

- 20 MHz: $346.7 \times 277.78 \times 10^3 \approx 96.3$ Mbps
- 40 MHz: $720.0 \times 277.78 \times 10^3 \approx 200.0$ Mbps
- 80 MHz: $1560.0 \times 277.78 \times 10^3 \approx 433.3$ Mbps
- 160 MHz: $3120.0 \times 277.78 \times 10^3 \approx 866.7$ Mbps

Numerical Example 1: Channel Bandwidth Impact IV

Step 4: With 2 spatial streams

- 20 MHz: $96.3 \times 2 = 192.6$ Mbps
- 40 MHz: $200.0 \times 2 = 400.0$ Mbps
- 80 MHz: $433.3 \times 2 = 866.7$ Mbps
- 160 MHz: $866.7 \times 2 = 1733.3$ Mbps

Final Answers:

- 1 20 MHz: **192.6 Mbps**
- 2 40 MHz: **400.0 Mbps**
- 3 80 MHz: **866.7 Mbps**
- 4 160 MHz: **1733.3 Mbps**

Observation:

- Increasing channel bandwidth increases PHY rate significantly.

Numerical Example 1: Channel Bandwidth Impact V

- The increase is slightly more than linear from 20 MHz to 80 MHz because wider channels use subcarriers more efficiently.
- From 80 MHz to 160 MHz, the PHY rate is approximately doubled.
- Note: 256-QAM is standard in 802.11ac, not in baseline 802.11n.

256-QAM: Pushing the Modulation Envelope I

- **802.11n:** Up to 64-QAM (6 bits/symbol)
- **802.11ac:** 256-QAM (8 bits/symbol)
- **Throughput Gain:** 256-QAM provides 33% more bits than 64-QAM

SNR Requirements:

Modulation	Bits/Symbol	Approx. SNR Required	Throughput Relative
BPSK	1	6 dB	1×
QPSK	2	9 dB	2×
16-QAM	4	16 dB	4×
64-QAM	6	22 dB	6×
256-QAM	8	28 dB	8×

Challenge: 256-QAM requires excellent SNR (28+ dB)

- Only works at short range (~30 ft) with clear line of sight
- Very sensitive to interference and multipath
- Beamforming helps achieve required SNR

Multi-User MIMO (MU-MIMO): Serving Multiple Clients Simultaneously

- **802.11n:** SU-MIMO (Single-User) - one client at a time
- **802.11ac:** MU-MIMO (Multi-User) - multiple clients simultaneously
- **Direction:** Downlink only (AP to clients) in Wave 2
- **Maximum:** 4 simultaneous clients (with 8×8 AP)

How it Works:

- 1 AP has multiple antennas (e.g., 8×8)
- 2 AP uses beamforming to create spatial separation
- 3 Sends different data to different clients at same time
- 4 Clients must support MU-MIMO and provide channel feedback

Analogy: AP becomes a switch instead of a hub

MU-MIMO vs SU-MIMO Comparison

Aspect	SU-MIMO (802.11n)	MU-MIMO (802.11ac Wave 2)
Clients Served	One at a time	Up to 4 simultaneously
Efficiency	Low with mixed clients	High with mixed clients
Client Requirements	Any MIMO client	MU-MIMO capable client
Beamforming	Optional, single target	Required, multiple targets
Complexity	Lower	Much higher (precoding)
Best For	Few clients, similar capabilities	Many clients, mixed capabilities

Key Benefit: Eliminates "rate anomaly" problem where slow clients slow down fast clients

Numerical Example 2: MU-MIMO Efficiency Gain I

Problem: Compare airtime usage for:

- SU-MIMO: Serving 3 clients sequentially
- MU-MIMO: Serving compatible clients simultaneously

Clients:

- Client A: 2×2 , 867 Mbps capable
- Client B: 2×2 , 867 Mbps capable
- Client C: 1×1 , 433 Mbps capable

Each client needs to download a 100 MB file.

Assumption: Practical throughput is 70% of PHY rate.

Step 1: SU-MIMO scenario

- Clients are served one at a time.

Numerical Example 2: MU-MIMO Efficiency Gain II

- Download time formula:

$$T = \frac{\text{File size in Mb}}{\text{PHY rate} \times 0.7}$$

- Since 100 MB = 800 Mb:

- Client A:

$$T_A = \frac{800}{867 \times 0.7} = \frac{800}{606.9} \approx 1.32 \text{ s}$$

- Client B:

$$T_B \approx 1.32 \text{ s}$$

- Client C:

$$T_C = \frac{800}{433 \times 0.7} = \frac{800}{303.1} \approx 2.64 \text{ s}$$

- Total SU-MIMO airtime:

$$T_{\text{SU}} = 1.32 + 1.32 + 2.64 = 5.28 \text{ s}$$

Numerical Example 2: MU-MIMO Efficiency Gain III

Step 2: MU-MIMO scenario

- A practical MU-MIMO grouping is to serve Clients A and B together, since both are similar 2×2 clients.
- Client C is served separately.
- Phase 1: A and B together

$$T_{A+B} = 1.32 \text{ s}$$

- Phase 2: C separately

$$T_C = 2.64 \text{ s}$$

- Total MU-MIMO airtime:

$$T_{\text{MU}} = 1.32 + 2.64 = 3.96 \text{ s}$$

Numerical Example 2: MU-MIMO Efficiency Gain IV

Step 3: Comparison

- SU-MIMO total time:

$$5.28 \text{ s}$$

- MU-MIMO total time:

$$3.96 \text{ s}$$

- Airtime reduction:

$$5.28 - 3.96 = 1.32 \text{ s}$$

- Relative improvement:

$$\frac{5.28}{3.96} \approx 1.33$$

Model Answer:

- SU-MIMO total time: **5.28 s**
- MU-MIMO total time: **3.96 s**

Numerical Example 2: MU-MIMO Efficiency Gain V

- **Efficiency gain:** MU-MIMO gives about $1.33\times$ faster completion than SU-MIMO in this case.

Observation:

- MU-MIMO works best when clients can be grouped efficiently.
- Similar high-rate clients benefit most from simultaneous transmission.
- A slower client may still need separate service, so gain is often less than the ideal $2\times$.

Beamforming Improvements: Explicit vs Implicit

- **802.11n Beamforming:**

- Mostly implicit (AP estimates channel)
- Vendor proprietary
- Limited interoperability

- **802.11ac Beamforming:**

- Explicit beamforming standard
- Client provides channel state information (CSI)
- AP computes precise beamforming weights
- Required for MU-MIMO

Beamforming Process:

- 1 AP sends Null Data Packet (NDP) announcement
- 2 AP sends NDP (sounding packet)
- 3 Client measures channel, sends feedback
- 4 AP computes beamforming weights
- 5 AP transmits with beamforming

Wave 1 vs Wave 2 802.11ac

Feature	Wave 1 (2013)	Wave 2 (2016)
Max Spatial Streams	3	4 (client), 8 (AP)
Channel Width	Up to 80 MHz	Up to 160 MHz (80+80)
MU-MIMO	No	Yes (downlink)
Maximum PHY Rate	1.3 Gbps (3×3, 80 MHz)	6.93 Gbps (8×8, 160 MHz)
Modulation	256-QAM	256-QAM
Beamforming	Explicit	Enhanced for MU-MIMO
Common Client Support	Good (most devices)	Limited (premium devices)

Market Reality: Most "802.11ac" devices are Wave 1. True Wave 2 benefits require both AP and clients to support.

Spatial Stream Scaling: Up to 8 Streams

- **802.11n:** Up to 4 spatial streams
- **802.11ac:** Up to 8 spatial streams (AP side)
- **Client Limitations:** Most clients 1×1, 2×2, or 3×3

Configuration	80 MHz PHY Rate	160 MHz PHY Rate	Typical Devices
1×1 (SISO)	433 Mbps	867 Mbps	Phones, IoT
2×2	867 Mbps	1.73 Gbps	Most laptops, tablets
3×3	1.3 Gbps	2.6 Gbps	Premium laptops
4×4	1.73 Gbps	3.47 Gbps	High-end APs
8×8	3.47 Gbps	6.93 Gbps	Enterprise APs

Note: 8×8 AP can serve multiple 2×2 clients simultaneously via MU-MIMO. Real-world example (YouTube): Four users, each with a 2×2 laptop, watching different YouTube videos. The 8×8 AP sends 2 streams to each laptop simultaneously. Each laptop gets roughly twice the speed of a 1×1 client, and all four watch without waiting for each other.

Numerical Example 3: Maximum 802.11ac Throughput I

Problem: Calculate maximum theoretical PHY rates for:

- 1 Wave 1: 3×3 MIMO, 80 MHz, 256-QAM
- 2 Wave 2: 4×4 SU-MIMO, 160 MHz, 256-QAM
- 3 Wave 2: 8×8 MU-MIMO serving $4 \times (2 \times 2)$ clients, 160 MHz

Then calculate practical TCP throughput assuming 75% efficiency.

Given:

- The given base PHY rates (433 Mbps for 80 MHz, 867 Mbps for 160 MHz) already include the modulation (256-QAM), coding rate (5/6), short guard interval, and the number of data subcarriers for that channel width.
- MU-MIMO efficiency factor: 80% (to account for precoding and feedback overhead)

Solution Steps:

Numerical Example 3: Maximum 802.11ac Throughput II

① Wave 1: 3×3 , 80 MHz

- PHY rate:

$$433 \times 3 = 1299 \text{ Mbps} \approx 1.30 \text{ Gbps}$$

- Practical TCP throughput:

$$1299 \times 0.75 = 974.25 \text{ Mbps} \approx 974 \text{ Mbps}$$

② Wave 2: 4×4 , 160 MHz

- PHY rate:

$$867 \times 4 = 3468 \text{ Mbps} \approx 3.47 \text{ Gbps}$$

- Practical TCP throughput:

$$3468 \times 0.75 = 2601 \text{ Mbps} \approx 2.60 \text{ Gbps}$$

③ Wave 2 MU-MIMO: 8×8 AP serving $4 \times (2 \times 2)$ clients

Numerical Example 3: Maximum 802.11ac Throughput III

- Each client PHY rate:

$$867 \times 2 = 1734 \text{ Mbps}$$

- Raw aggregate PHY rate:

$$1734 \times 4 = 6936 \text{ Mbps} \approx 6.94 \text{ Gbps}$$

- Effective aggregate PHY after MU-MIMO overhead:

$$6936 \times 0.8 = 5548.8 \text{ Mbps} \approx 5.55 \text{ Gbps}$$

- Practical aggregate TCP throughput:

$$5548.8 \times 0.75 = 4161.6 \text{ Mbps} \approx 4.16 \text{ Gbps}$$

Model Answer:

- ① Wave 1: **1.30 Gbps PHY, 974 Mbps TCP**

Numerical Example 3: Maximum 802.11ac Throughput IV

- ② Wave 2 SU-MIMO: **3.47 Gbps PHY, 2.60 Gbps TCP**
- ③ Wave 2 MU-MIMO:
 - **6.94 Gbps raw aggregate PHY**
 - **5.55 Gbps effective aggregate PHY**
 - **4.16 Gbps aggregate TCP**

Note:

- The MU-MIMO throughput is aggregate across 4 clients.
- Average TCP throughput per client is approximately:

$$\frac{4.16}{4} \approx 1.04 \text{ Gbps}$$

- Actual per-client throughput may differ depending on scheduling, channel quality, and overheads.

Backward Compatibility with 802.11n/a

- **802.11ac operates only in 5 GHz**
- **Backward compatible with:**
 - 802.11a (OFDM, 20 MHz)
 - 802.11n (HT, 20/40 MHz)
- **Protection Mechanisms:**
 - VHT (Very High Throughput) protection for legacy devices
 - Similar to 802.11n mixed mode
 - Uses legacy rates for control frames
- **Impact:** Legacy devices reduce 802.11ac performance

Recommendation: Separate SSIDs for 802.11ac-only networks

Real-World 802.11ac Performance

Client Device	Typical Configuration	Theoretical Max	Real-World TCP
Smartphone	1×1 or 2×2, 80 MHz	433-867 Mbps	200-400 Mbps
Tablet	1×1 or 2×2, 80 MHz	433-867 Mbps	200-400 Mbps
Laptop	2×2, 80 MHz	867 Mbps	400-500 Mbps
Premium Laptop	3×3, 80 MHz	1.3 Gbps	600-800 Mbps
Desktop Adapter	3×3, 160 MHz	1.3-2.6 Gbps	700-1500 Mbps

Factors Reducing Performance:

- 1 Distance (256-QAM requires close proximity)
- 2 Interference (especially with wide channels)
- 3 Legacy devices on network
- 4 Client limitations (most are 2×2 or less)

1 Channel Planning:

- 80 MHz channels reduce available channels significantly
- 160 MHz even more so - often impractical in dense deployments
- Consider 20/40 MHz for high-density, 80 MHz for low-density

2 AP Placement:

- 256-QAM requires good SNR - place APs closer together
- MU-MIMO benefits from clients in different spatial locations

3 Client Support:

- Most clients support 80 MHz, 2×2
- Few support 160 MHz or 3×3 +
- Even fewer support MU-MIMO

4 Uplink Speeds:

- Switches need multi-gigabit ports (1 Gbps insufficient)
- Consider 2.5G, 5G, or 10G Ethernet for APs

- **Target Wake Time (TWT):** Introduced in 802.11ah, optional in 802.11ac
- **Multi-User Beamforming:** More efficient than single-user
- **Spatial Reuse:** Higher rates = less airtime = less radio on-time

Battery Impact:

- Higher rates can actually save power (finish transfers faster)
- But 256-QAM requires more processing power
- Net effect: Mixed, but generally positive for battery life

Limitations and Challenges of 802.11ac

- ❶ **5 GHz Only:** Requires dual-band APs for 2.4 GHz compatibility
- ❷ **Wide Channel Trade-off:** Fewer non-overlapping channels
- ❸ **MU-MIMO Complexity:** Difficult to implement, limited client support
- ❹ **256-QAM Range:** Only works at short distances
- ❺ **Real-World vs Theoretical:** Most clients achieve far less than maximum
- ❻ **Uplink Bottleneck:** Still SU-MIMO on uplink (client to AP)
- ❼ **Interference Sensitivity:** Wider channels catch more interference

These limitations led to 802.11ax (Wi-Fi 6)

802.11ac vs 802.11n Performance Comparison

Metric	802.11n (3×3, 40 MHz)	802.11ac Wave 1 (3×3, 80 MHz)	Improvement
Max PHY Rate	450 Mbps	1300 Mbps	2.9×
Practical TCP	200-250 Mbps	600-800 Mbps	3×
Channel Width	40 MHz max	80 MHz	2×
Modulation	64-QAM max	256-QAM	1.33×
Efficiency	Medium (aggregation)	High (better aggregation)	1.2×
Real-World Gain	Baseline	2-3× faster	Significant

Note: 2-3× real-world improvement, with potential for more with MU-MIMO and ideal conditions

MU-MIMO Implementation Challenges

- **Channel Feedback:** Clients must provide accurate CSI
- **Client Mobility:** Beamforming weights become stale quickly
- **Client Selection:** Which clients to group together?
- **Buffer Management:** AP must buffer frames for multiple clients
- **Scheduling Complexity:** When to use MU-MIMO vs SU-MIMO?

Real-World MU-MIMO Performance:

- Often provides 20-40% capacity gain (not the theoretical $2-4\times$)
- Requires compatible clients (most aren't)
- Works best with stationary clients

Summary: 802.11ac Key Innovations

- ❶ **5 GHz Exclusive:** Clean spectrum for high performance
- ❷ **Wider Channels:** 80 and 160 MHz (2-4 $\times$ bandwidth)
- ❸ **256-QAM:** 33% more bits per symbol (requires high SNR)
- ❹ **MU-MIMO:** Serve multiple clients simultaneously (downlink)
- ❺ **Explicit Beamforming:** Standardized, required for MU-MIMO
- ❻ **More Spatial Streams:** Up to 8 (AP side)
- ❼ **Wave 2 Enhancements:** 160 MHz, MU-MIMO, more streams

Overall: 802.11ac delivered the first true gigabit Wi-Fi, but with limitations that led to the next generation.

- **Required Reading:**

- Textbook (Gast): Chapter 9 - "802.11ac: Very High Throughput"
- Review MIMO and OFDM concepts

- **Optional Reading:**

- IEEE 802.11ac-2013 standard (selected sections)
- MU-MIMO technical papers

- **Next Lecture (Lecture 10): 802.11ax (Wi-Fi 6 & 6E)**

- OFDMA for multi-user efficiency
- Uplink MU-MIMO
- BSS Coloring for spatial reuse
- Target Wake Time for power efficiency
- 6 GHz band introduction (Wi-Fi 6E)
- 1024-QAM modulation

Review Questions

- 1 Why is 802.11ac limited to 5 GHz band?
- 2 Calculate: 2×2 client with 80 MHz vs 160 MHz channel - throughput difference?
- 3 What SNR is required for 256-QAM vs 64-QAM?
- 4 How does MU-MIMO solve the "rate anomaly" problem?
- 5 What are the practical limitations of 160 MHz channels?
- 6 Why is explicit beamforming required for MU-MIMO?
- 7 Compare Wave 1 vs Wave 2 802.11ac features.
- 8 Why do most clients achieve far less than maximum 802.11ac speeds?

Discussion Question: "802.11ac introduced 160 MHz channels, but they're rarely used in enterprise deployments. Why did this feature fail to gain widespread adoption, and what does this tell us about the trade-off between channel width and network density?"

Exercise: 802.11ac Performance Analysis

- **Objective:** Measure and analyze 802.11ac performance characteristics
- **Tools:** 802.11ac capable devices, Wireshark, iPerf, Wi-Fi analyzer
- **Tasks:**
 - 1 Measure throughput at different distances (256-QAM falloff)
 - 2 Compare 80 MHz vs 40 MHz channel performance
 - 3 Test MU-MIMO with multiple clients (if supported)
 - 4 Analyze beamforming behavior
 - 5 Measure impact of legacy devices on network
- **Deliverable:** Performance analysis report with recommendations

Appendix: 802.11ac MCS Tables (80 MHz) I

MCS Index	Modulation	Coding Rate	Spatial Streams	PHY Rate (Mbps) GI=800ns	SNR Required GI=400ns
0 ~6 dB	BPSK	1/2	1	293	325
1 ~9 dB	QPSK	1/2	1	585	650
2 ~11 dB	QPSK	3/4	1	878	975
3 ~16 dB	16-QAM	1/2	1	1170	1300
4 ~18 dB	16-QAM	3/4	1	1755	1950
5 ~22 dB	64-QAM	2/3	1	2340	2600
6 ~24 dB	64-QAM	3/4	1	2633	2925
7 ~25 dB	64-QAM	5/6	1	2925	3250
8 ~28 dB	256-QAM	3/4	1	3510	3900
9 ~30 dB	256-QAM	5/6	1	3900	4333

Note: For multiple spatial streams, multiply by number of streams. For 160 MHz, approximately double these rates.

Appendix: 802.11ac Channel Availability (US)

Band	Frequency Range	Channels (20 MHz)	80 MHz Channels	160 MHz Channels
UNII-1	5.150-5.250 GHz	36, 40, 44, 48	1 (36-48)	0
UNII-2A	5.250-5.350 GHz	52, 56, 60, 64	1 (52-64)	0
UNII-2C	5.470-5.725 GHz	100-140 (8 channels)	2 (100-112, 116-128)	1 (100-128)
UNII-3	5.725-5.850 GHz	149, 153, 157, 161	1 (149-161)	0
Total		20	5	1

DFS Required: UNII-2 bands require radar avoidance

- Causes occasional channel changes
- Some consumer devices disable DFS channels

Practical: 2-3 usable 80 MHz channels in most locations

Appendix: MU-MIMO Grouping Example

Scenario: 8×8 AP with clients:

- Client A: 2×2 , good SNR
- Client B: 2×2 , good SNR
- Client C: 1×1 , moderate SNR
- Client D: 3×3 , excellent SNR
- Client E: 2×2 , poor SNR

Possible MU-MIMO Groups:

- ① **Group 1 (good spatial separation):** A + B (both 2×2 , good SNR)
- ② **Group 2 (different capabilities):** C + D ($1 \times 1 + 3 \times 3$) - challenging
- ③ **Poor Group:** E with anyone (poor SNR affects whole group)

Scheduling Algorithm Considerations:

- Similar channel conditions
- Similar traffic requirements
- Good spatial separation (different angles from AP)
- Similar capabilities (stream counts)

Appendix: 802.11ac Timeline and Adoption

- **2008:** Work begins on Very High Throughput (VHT) task group
- **2011:** First draft products (pre-standard)
- **2013:** Wave 1 standard ratified
- **2014:** Apple includes 802.11ac in MacBooks
- **2015:** Most new laptops include 802.11ac
- **2016:** Wave 2 enhancements ratified
- **2017:** First Wave 2 consumer products
- **2018:** 802.11ac dominant in market
- **2020:** Still widely used alongside 802.11ax

Adoption Rate: Faster than 802.11n due to clear benefits and 5 GHz availability

Appendix: Power Consumption Analysis

802.11ac vs 802.11n Power Consumption:

- **Higher rates:** Finish transfers faster → radio off sooner
- **256-QAM:** More processing power required
- **MU-MIMO:** More complex processing but serves multiple clients
- **Wide channels:** More bandwidth but similar base power

Net Effect:

- **Bulk transfers:** Lower energy per bit (more efficient)
- **Idle/short bursts:** Similar or slightly higher power
- **Overall:** Positive for battery life with typical usage

Example: Downloading 1 GB file:

- 802.11n (200 Mbps): 40 seconds at high power
- 802.11ac (500 Mbps): 16 seconds at high power
- 802.11ac uses less total energy despite higher peak power

Thank you!